№T5.

$\xi \sim R[\theta, 2\theta]$, $\exists\ \vec{X_n}$

$\Rightarrow \xi \sim \rho(x,\theta) = \frac{1}{\theta}\{(\theta, 2\theta)\}$

а), в) Найдём $\tilde{\theta}$:

О.М.М. Возьмём $\alpha_1$:

$$\alpha_1 = \int_{-\infty}^{+\infty} x\rho\, dx = \int_{\theta}^{2\theta} \frac{x}{\theta}\, dx = \frac{3}{2}\theta$$

$$\alpha_1 = \tilde{\alpha}_1 = \bar{X} \Rightarrow \frac{3}{2}\theta = \bar{X} \Rightarrow \tilde{\theta} = \frac{2}{3}\bar{X} \text{ – О.М.М.}$$

Проверим несмещ.:

$$M[\tilde{\theta}] = \frac{2}{3}M[\bar{X}] = \frac{2}{3}M[\xi] = \theta \Rightarrow \tilde{\theta} \text{ – несмещ.}$$

Проверим состоятельность:

$$D[\tilde{\theta}] = \frac{4}{9}D[\bar{X}] = \frac{4}{9n}\underbrace{D[\xi]}_{\mu_2} \circleddash$$

$$\alpha_2 = \int_{\theta}^{2\theta} \frac{x^2}{\theta}\, dx = \frac{8t^3}{3t} - \frac{t^3}{3t} = \frac{7}{3}t^3 \Rightarrow \mu_2 = \alpha_2 - \alpha_1^2 = \frac{\theta^2}{12}$$

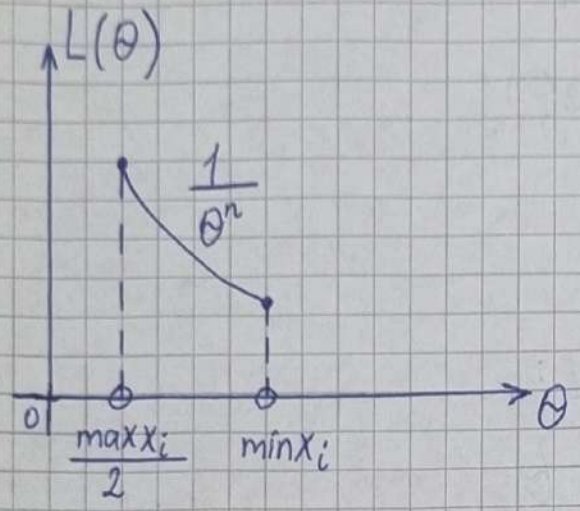
$$\circleddash \frac{4}{9n}\cdot\frac{\theta^2}{12} = \frac{\theta^2}{27n} \xrightarrow[n\to+\infty]{} 0 \Rightarrow \tilde{\theta} \text{ – состоят.}$$

О.М.П.

Функция правдоподобия:

$$L(\theta) = \prod_{i=1}^{n} \rho(x_i, \theta) = \frac{1}{\theta^n}\left\{\forall i \hookrightarrow \begin{matrix} \min x_i \geq \theta;\ i=1,\dots,n \\ \max x_i \leq 2\theta;\ i=1,\dots,n \end{matrix}\right\} =$$

$$= \frac{1}{\theta^n} \left\{ \frac{\max X_i}{2} \le \theta \le \min X_i \right\}$$




$$\Rightarrow \max: \ \tilde{\theta} = \frac{\max X_i}{2}$$

О.М.П.

Проверим несмещ.:

$$M[\tilde{\theta}] = \frac{1}{2} M[\max X_i] = \frac{1}{2} \int_{\theta}^{2\theta} t \cdot \frac{n}{\theta} \left( \frac{t}{\theta} - 1 \right)^{n-1} dt =$$

$$= \left\{ \begin{array}{l} u = \frac{t}{\theta} \\ du = \frac{1}{\theta} dt \\ t = u\theta \\ dt = \theta du \end{array} \right\} = \frac{\theta}{2} \int_1^2 nu(u-1)^{n-1} du = \frac{\theta}{2} \cdot \tilde{I} \ \textcircled{=}$$

$$\int u\,d\delta = u\delta - \int \delta\,du$$

$$\tilde{I} = \int_1^2 u \cdot \underbrace{n(u-1)^{n-1} du}_{d((u-1)^n)} = u(u-1)^n \Big|_1^2 - \int_1^2 (u-1)^n du = 2 + \frac{1^{n+1}}{n+1}$$

$$\textcircled{=} \ \theta - \frac{\theta}{2(n+1)} = \frac{\theta(2n+1)}{2n+2} \Rightarrow \tilde{\theta} \text{ - смещ.}$$

Тогда

$$\tilde{\theta}' = \frac{2n+2}{2n+1} \cdot \frac{\max X_i}{2} = \frac{n+1}{2n+1} \max X_i \text{ - несмещ.}$$

Проверим оптимальность:

$$D[\tilde{\theta}] = M[\tilde{\theta}^2] - M^2[\tilde{\theta}]$$

$$M[\tilde{\theta}^2] = \frac{1}{4}\int_{\theta}^{2\theta} t^2 \frac{n}{\theta}\left(\frac{t}{\theta}-1\right)^{n-1} dt = \left\{ u = \frac{t}{\theta} \right\} =$$

$$= \frac{\theta^2}{4}\int_1^2 n\cdot u^2 (u-1)^{n-1} du = \frac{\theta^2}{4}\cdot \tilde{\tilde{I}}$$

$$\tilde{\tilde{I}} = \int_1^2 n\cdot u^2 (u-1)^{n-1} du = u^2(u-1)^n\Big|_1^2 - 2\int_1^2 u(u-1)^n du =$$

$$= 4 - \frac{2}{n+1}\int_1^2 (n+1)u(u-1)^n du \ominus$$

$$\tilde{\tilde{\tilde{I}}} = \int_1^2 (n+1)u(u-1)^n du = u(u-1)^{n+1}\Big|_1^2 - \frac{(u-1)^{n+2}}{n+2}\Big|_1^2 =$$

$$= 2 - \frac{1^{n+2}}{n+2} = 2 - \frac{1}{n+2}$$

$$\ominus\ 4 - \frac{2}{n+1}\left(2 - \frac{1}{n+2}\right) = \frac{2(2n^2+4n+1)}{(n+2)(n+1)}$$

$$\Rightarrow M[\tilde{\theta}^2] = \frac{\theta^2}{4}\cdot\frac{2(2n^2+4n+1)}{(n+2)(n+1)} = \frac{2n^2+4n+1}{2(n+2)(n+1)}\cdot\theta^2$$

Тогда

$$D[\tilde{\theta}] = \theta^2\left(\frac{2n^2+4n+1}{2(n+2)(n+1)} - \left(\frac{2n+1}{2n+2}\right)^2\right) = \theta^2\cdot\frac{n}{4(n+1)^2(n+2)}$$

(если смотреть состоят. $\tilde{\theta}$, то по опр.)

$$D[\tilde{\theta}'] = 4\cdot\frac{(n+1)^2}{(2n+1)^2}\cdot\theta^2\cdot\frac{n}{4(n+1)^2(n+2)} \xrightarrow[n\to\infty]{} 0$$

$\Rightarrow \tilde{\theta}'$ - состоят.

c) $D[\underset{\text{ОММ}}{\tilde{\theta}}] = \dfrac{\theta^2}{27n} \geqslant \dfrac{n\theta^2}{(n+2)(2n+1)^2} = D[\underset{\text{ОМП}}{\tilde{\theta}'}]$

(при $n \geqslant 3$)

$\Rightarrow$ $\underset{\text{(исправл.)}}{\text{ОМП}}$ эффективнее ОММ

d) Пусть $f(\vec{X_n}, \theta) = \dfrac{\max\limits_{i=1,\dots,n} X_i}{\theta} - 1$

Тогда $P(f < t) = P(\max X_i < \theta t + \theta) =$

$= (F(\theta t + \theta))^n$, где $F(x) = \begin{cases} 0, & x < \theta \\ \frac{x}{\theta} - 1, & \theta \leq x \leq 2\theta \\ 1, & x > 2\theta \end{cases}$

т.о. $P(f < t) = \begin{cases} 0, & t < 0 \\ t^n, & 0 \leq t \leq 1 \\ 1, & t > 1 \end{cases}$

$t_1 : F(t_1) = \dfrac{1-\beta}{2}$

$t_2 : F(t_2) = \dfrac{1+\beta}{2}$

$\Rightarrow t_{1,2} = \sqrt[n]{\dfrac{1 \mp \beta}{2}}$

Тогда $P\left(t_1 < \dfrac{\max X_i}{\theta} - 1 < t_2\right) = \beta$

т.е. $t_1 + 1 < \dfrac{\max X_i}{\theta} < t_2 + 1$

$$\Rightarrow P\left(\frac{\max X_i}{1 + \sqrt[n]{\frac{1+\beta}{2}}} < \theta < \frac{\max X_i}{1 + \sqrt[n]{\frac{1+\beta}{2}}}\right) = \beta$$

точный доверительный интервал.

е) ОММ: $\tilde{\theta} = \frac{2}{3}\bar{X} = g(\tilde{\alpha}_1)$

$g(\alpha_1) = \theta$; $\vec{\nabla} g = g' = \frac{2}{3}$; $K_{11} = \alpha_2 - \alpha_1^2$

$$\frac{\sqrt{n}(\tilde{\theta} - \theta)}{\sqrt{\left(\frac{2}{3}\right)^2 (\tilde{\alpha}_2 - \tilde{\alpha}_1^2)}} = \frac{3n(\tilde{\theta} - \theta)}{2\sqrt{S^2(n-1)}} \rightsquigarrow N(0,1),$$ где

$$S^2 = \frac{n}{n-1} \cdot \tilde{\mu}_2$$

т.к. норм. распр.: $t_{1,2} = \mp 1{,}96$

$$P\left(t_1 < \frac{3n(\tilde{\theta} - \theta)}{2\sqrt{S^2(n-1)}} < t_2\right) = \beta$$

тогда $$\frac{2t_1\sqrt{S^2(n-1)}}{3n} < \tilde{\theta} - \theta < \frac{2t_2\sqrt{S^2(n-1)}}{3n}$$

$$\Rightarrow P\left(\frac{2}{3}\bar{X} - \frac{3{,}92\sqrt{S^2(n-1)}}{3n} < \theta < \frac{2}{3}\bar{X} + \frac{3{,}92\sqrt{S^2(n-1)}}{3n}\right) \beta$$

асимптотический доверительный интервал ОММ